DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

M.Tech/Ph.D MID-II SEMESTER: 09th OctNov, 2022

Course (Subject) : **Advanced Optimization Techniques** Course Code : **AM-603**

Duration : **01 hr** Max Marks : **10**

Note: Attempt all the **two** questions from the following. All questions carry equal marks.

1. a). Explain about Critical Path Method and PERT

b) Use Dynamic programming to solve the L.P.P,

$$\text{Maxmize } Z = 8x_1 + 7x_2$$

S.to.

$$2x_1 + x_2 \le 8$$

$$5x_1 + 2x_2 \le 15$$

$$x_1,\ x_2 \ge 0$$

2. A machine operator processes five types of items in his machine each week and must choose a sequence for them. The set up cost per change depends on the item presently on the machine and the set up to be made according to the following table:

	A	B	C	D	E
A	-	4	7	3	4
B	4	-	6	3	4
C	7	6	-	7	5
D	3	3	7	-	7
E	4	4	5	7	-

If he processes each type of item once and only once each week, how should he sequence the items on his machine in order to minimize the total set up cost?